

Review retrospective:
*the mixed odd-moment vanishing — a clean file, an empty
bucket*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/TwoD/KthKthOddMoments.lean`: the two parity-completion theorems stating that a mixed 2D moment $\langle x^a y^b \rangle_{2D}$ on the separable monomial potential vanishes whenever at least one of a, b is odd. Result: **a clean fidelity pass with no edits** — the file carries no dead `open`, no dead import, and no no-op tactic. The reliably-safe simplification bucket was empty, and that is the result.

1 Setting

This is the odd-moment companion to the `kthkth` lift trio (partition, numerator, moment): where the trio computes the *even*-moment asymptotics, this file records that the *odd* ones vanish identically. It sits on the same factorisation root (`AddSeparable`) and 1D base (`MonomialPotential`).

2 What was reviewed

Two theorems: `gibbsExpectation_kthKth_odd_pow_pow_eq_zero` ($\langle x^{2j_1+1} y^n \rangle_{2D} = 0$ for any n) and `gibbsExpectation_kthKth_pow_odd_pow_eq_zero` ($\langle x^m y^{2j_2+1} \rangle_{2D} = 0$ for any m). Each factorises the 2D expectation into a product of two 1D ones and kills the odd factor via `gibbsExpectation_kthPotential_odd`; the product is zero by `zero_mul/mul_zero`.

3 Fidelity / soundness findings

Sound; no mismatch. The load-bearing check is that the *non-odd* factor needs only *integrability* — supplied by `kth_integrable_pow_pot` for *every* natural exponent, not just even ones — so each theorem genuinely holds for arbitrary n (resp. m), and the two together cover every “at least one odd” case, including odd–odd (apply whichever side’s exponent is odd; if both, either). The per-theorem docstrings’ “regardless of the parity of n/m , including odd–odd” is exactly justified.

One report-only prose nit: the module-level docstring states the vanishing without restating the $t > 0$ hypothesis that the formal theorems carry (and the proofs use, via partition positivity). The $\langle \cdot \rangle_{2D,t}$ notation carries the t , and the result *is* true for all $t > 0$, so this is terse rather than wrong — left as is, consistent with the line drawn on the `AddSeparable` docstring-completeness notes.

4 Simplifications applied

None — the bucket was empty. Unlike its three trio siblings, this file opens *only* `MeasureTheory` (load-bearing for bare `Integrable`); there is no dead `open Real` to trim, both imports are used, the (correctly absent) `Asymptotics` import is not present to remove, and the final `simp` does real work (`zero_mul/mul_zero`). The reviewer’s only suggestions were proof-golf (`simp`-shorten the $n = 0$ integrability blocks; compress the closing `rw;simp`), declined as out of the signature-preserving dead-code safe class. No worktree was created.

5 Disagreements and open questions

None.

6 Lessons

- **An empty reliably-safe bucket is a valid — and common — outcome.** A file written without the accreted dead `open Real` of its siblings yields a clean fidelity pass and nothing to land; manufacturing a mutation to “show work” would violate the review’s safety contract. The result is the clean pass plus the report-only prose nit.
- **The sibling-prediction cuts both ways.** The trio legs all carried a dead `open Real`; this companion did not. A copy-paste family tends to share dead code, but membership is not destiny — the per-file check (here: the open line literally reads `open MeasureTheory`) still governs.

7 Follow-ups

- The $(2, 3)$ `QuarticSextic*` specialisations (`Partition/Numerator/Moment/OddMoments` and the `QuarticSextic/QuarticSexticMoments` cores) remain — some near-certain dead-`open Real` repeats, some (like this one) possibly already clean.
- Proof-golf forward tide (shared across the whole `kthkth` family): the `simp` integrability shortenings.